

Problem 2

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Problem 1 Complete steps (i)-(vii) below in order to estimate the following values using linear approximation:

(a) $\cos\left(\frac{31\pi}{180}\right)$

(b) $\sqrt[3]{8.13}$

- (i) Identify the function, $f(x)$.
- (ii) Find the nearby value where the function can be easily calculated, $x = a$.
- (iii) Find $\Delta x = dx$.
- (iv) Find the linear approximation, $L(x)$. [2]
- (v) Compute the approximate value of the expression using the linear approximation.
- (vi) Compare the approximated value to the value given by your calculator.
- (vii) Compare dy and Δy using the value given by your calculator. [2]

Explanation. (a) $\cos\left(\frac{31\pi}{180}\right)$

- (i) $f(x) = \cos x$
- (ii) $a = \frac{30\pi}{180} = \frac{\pi}{6}$
- (iii) $\Delta x = \frac{31\pi}{180} - \frac{\pi}{6} = \frac{\pi}{180}$
- (iv)

$$\begin{aligned} L(x) &= f\left(\frac{\pi}{6}\right) + f'\left(\frac{\pi}{6}\right)\left(x - \frac{\pi}{6}\right) \\ &= \cos\left(\frac{\pi}{6}\right) - \sin\left(\frac{\pi}{6}\right)\left(x - \frac{\pi}{6}\right) \\ &= \frac{\sqrt{3}}{2} - \frac{1}{2}\left(x - \frac{\pi}{6}\right) \end{aligned}$$

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(v)

$$\begin{aligned} L\left(\frac{31\pi}{180}\right) &= \frac{\sqrt{3}}{2} - \frac{1}{2}\left(\frac{31\pi}{180} - \frac{\pi}{6}\right) \\ &= \frac{\sqrt{3}}{2} - \frac{1}{2}\left(\frac{\pi}{180}\right) \\ &= \frac{1}{2}\left(\sqrt{3} - \frac{\pi}{180}\right) \\ &\approx 0.857299 \end{aligned}$$

(vi) $\cos\left(\frac{31\pi}{180}\right) \approx 0.857167$

(vii)

$$\begin{aligned} dy &= L\left(\frac{31\pi}{180}\right) - L\left(\frac{\pi}{6}\right) \approx -0.008727 \\ \Delta y &= \cos\left(\frac{31\pi}{180}\right) - \cos\left(\frac{\pi}{6}\right) \approx -0.008858 \end{aligned}$$

(b) $\sqrt[3]{8.13}$

(i) $f(x) = \sqrt[3]{x}$.

(ii) $a = 8$.

(iii) $\Delta x = 8.13 - 8 = 0.13$.

(iv)

$$\begin{aligned} L(x) &= f(8) + f'(8)(x - 8) \\ &= \sqrt[3]{8} + \frac{1}{3(\sqrt[3]{8})^2}(x - 8) \\ &= 2 + \frac{1}{12}(x - 8) \end{aligned}$$

(v)

$$\begin{aligned} L(8.13) &= 2 + \frac{1}{12}(8.13 - 8) \\ &= 2 + \left(\frac{1}{12}\right)\left(\frac{13}{100}\right) \\ &= 2 + \frac{13}{1200} = \frac{2413}{1200} \\ &\approx 2.010833 \end{aligned}$$

(vi) $\sqrt[3]{8.13} \approx 2.010775$.

(vii)

$$\begin{aligned} dy &= L(8.13) - L(8) \approx 0.010833 \\ \Delta y &= \sqrt[3]{8.13} - \sqrt[3]{8} \approx 0.010775 \end{aligned}$$